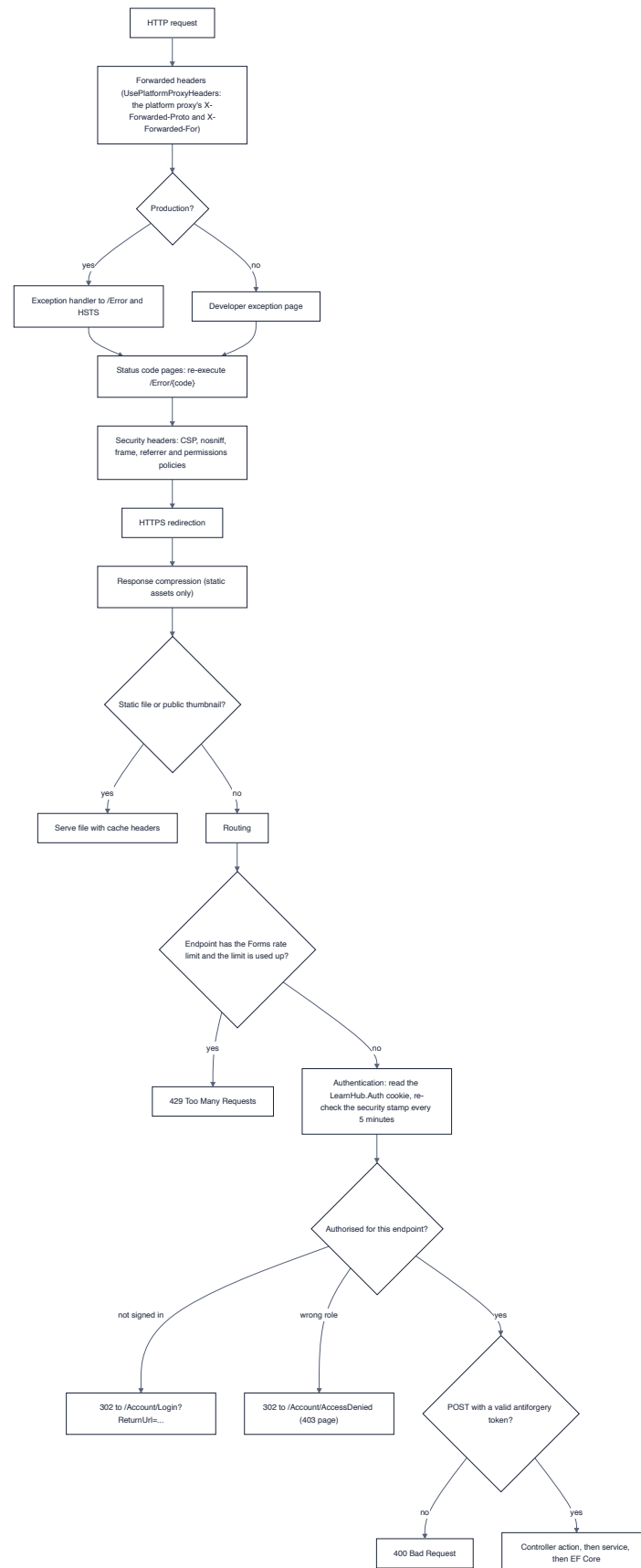


Flowcharts

Each flowchart follows the real code path. File paths are relative to `src/LearnHub/`.

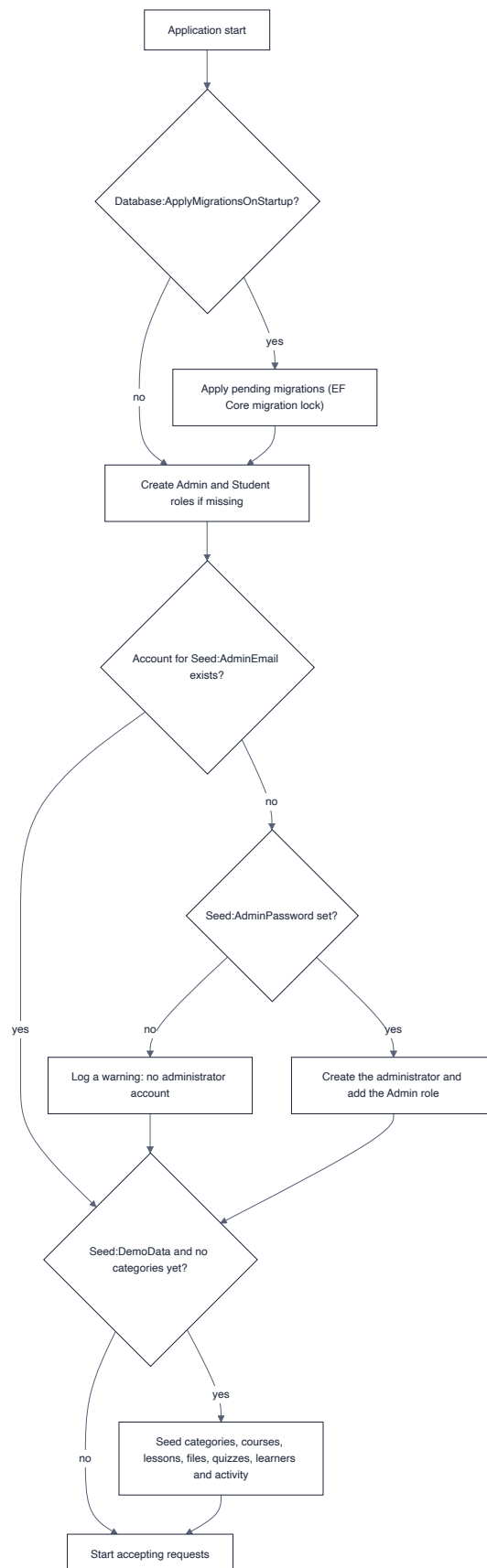
1. Request pipeline

Every request passes through the middleware configured in `Program.cs`. Railway terminates TLS at its edge proxy and forwards plain HTTP with `X-Forwarded-Proto` and `X-Forwarded-For`, so `UsePlatformProxyHeaders()` runs first and turns those headers into `Request.IsHttps` and the real client address.



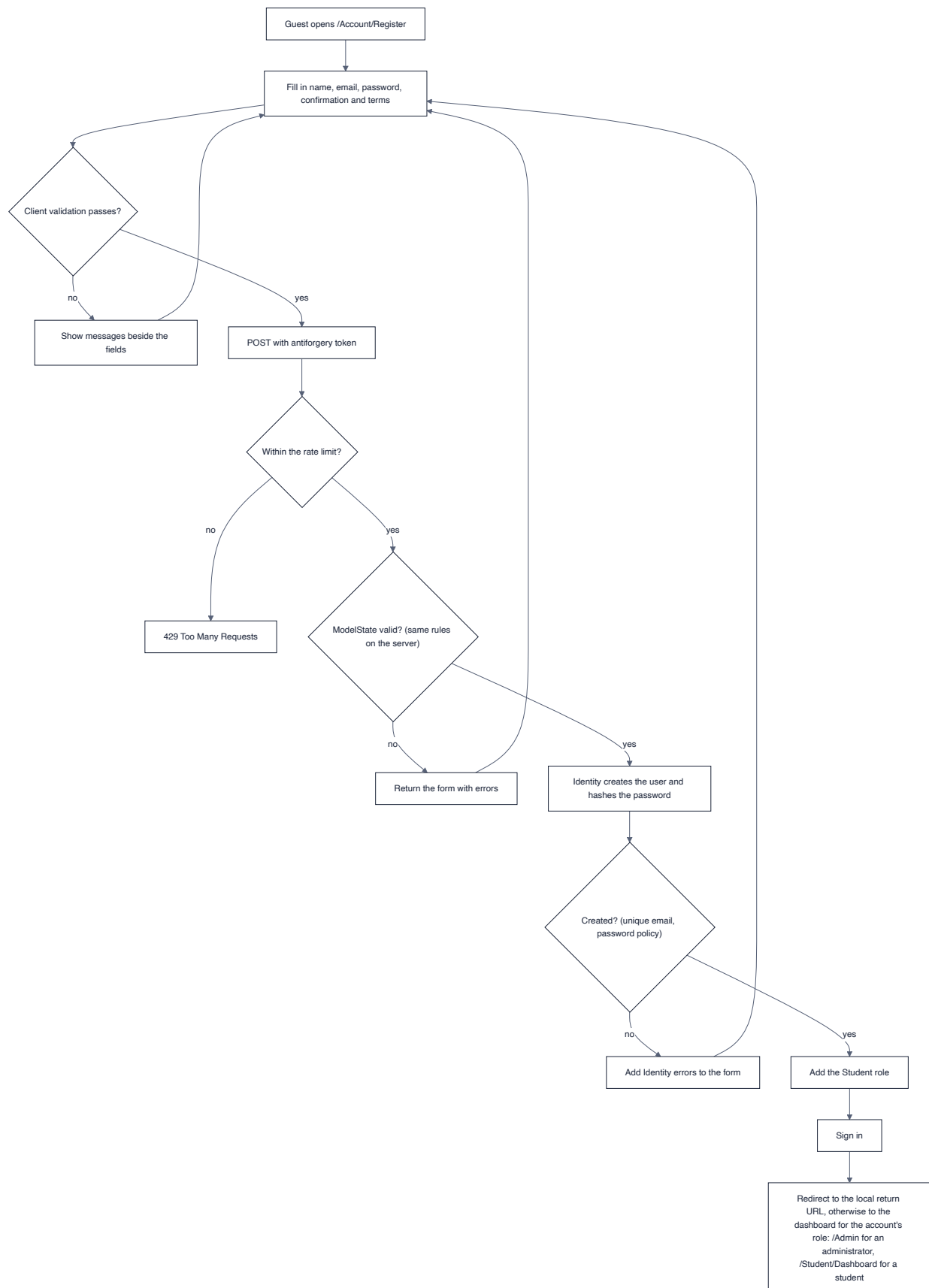
2. Start-up: database initialisation

`Data/Seed/DatabaseInitializer.cs` runs once before the application starts listening.



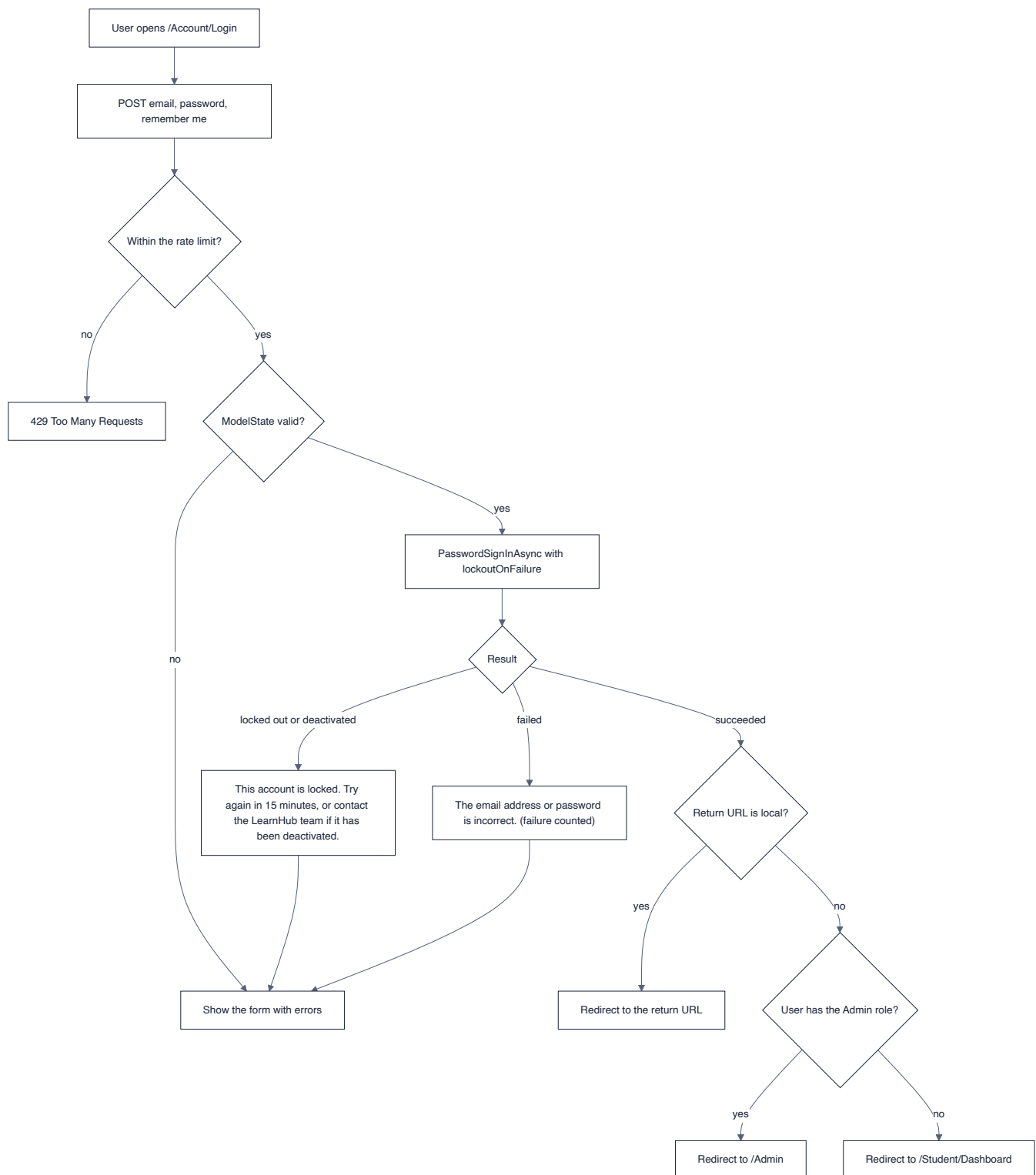
3. Registration

Controllers/AccountController.cs (Register) and ViewModels/Account/AccountViewModels.cs .



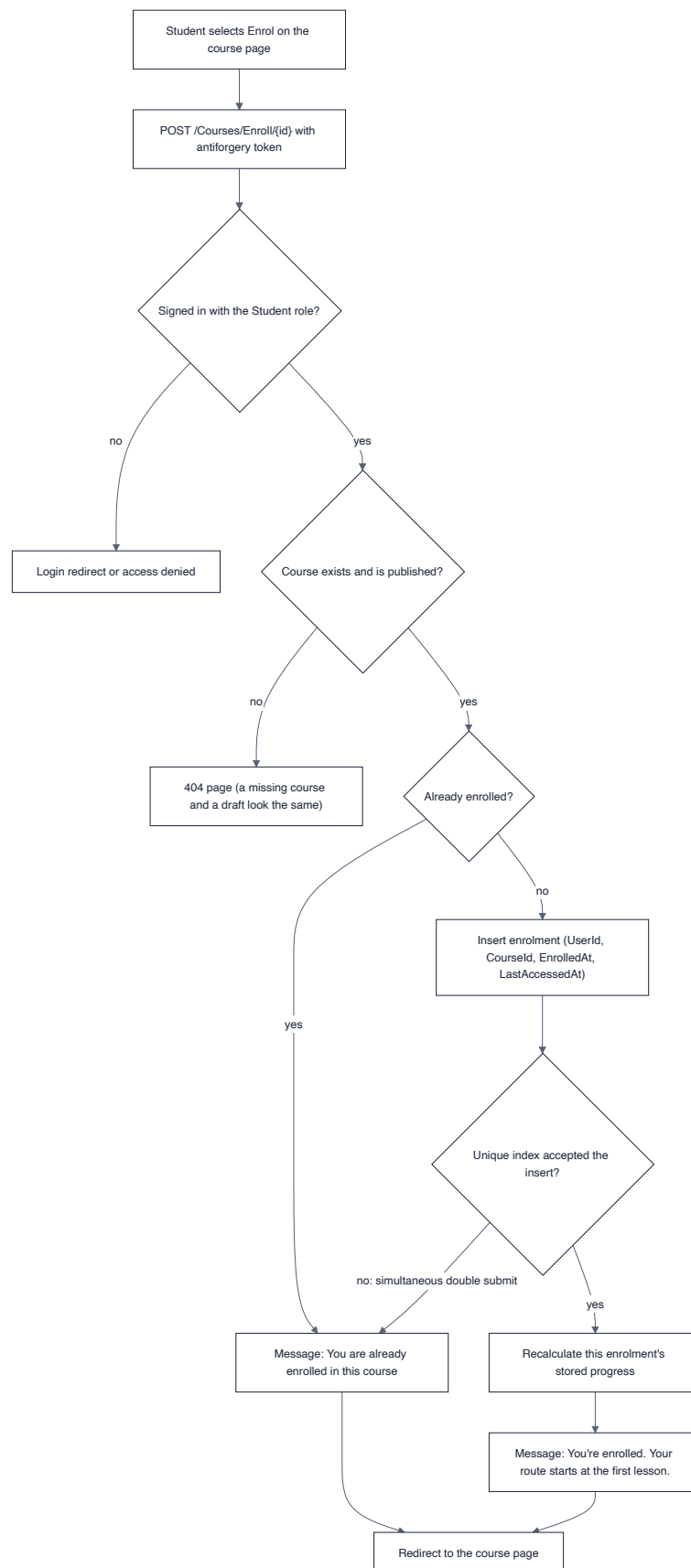
4. Login

`Controllers/AccountController.cs` (Login). Identity options: 5 failed attempts lock the account for 15 minutes.



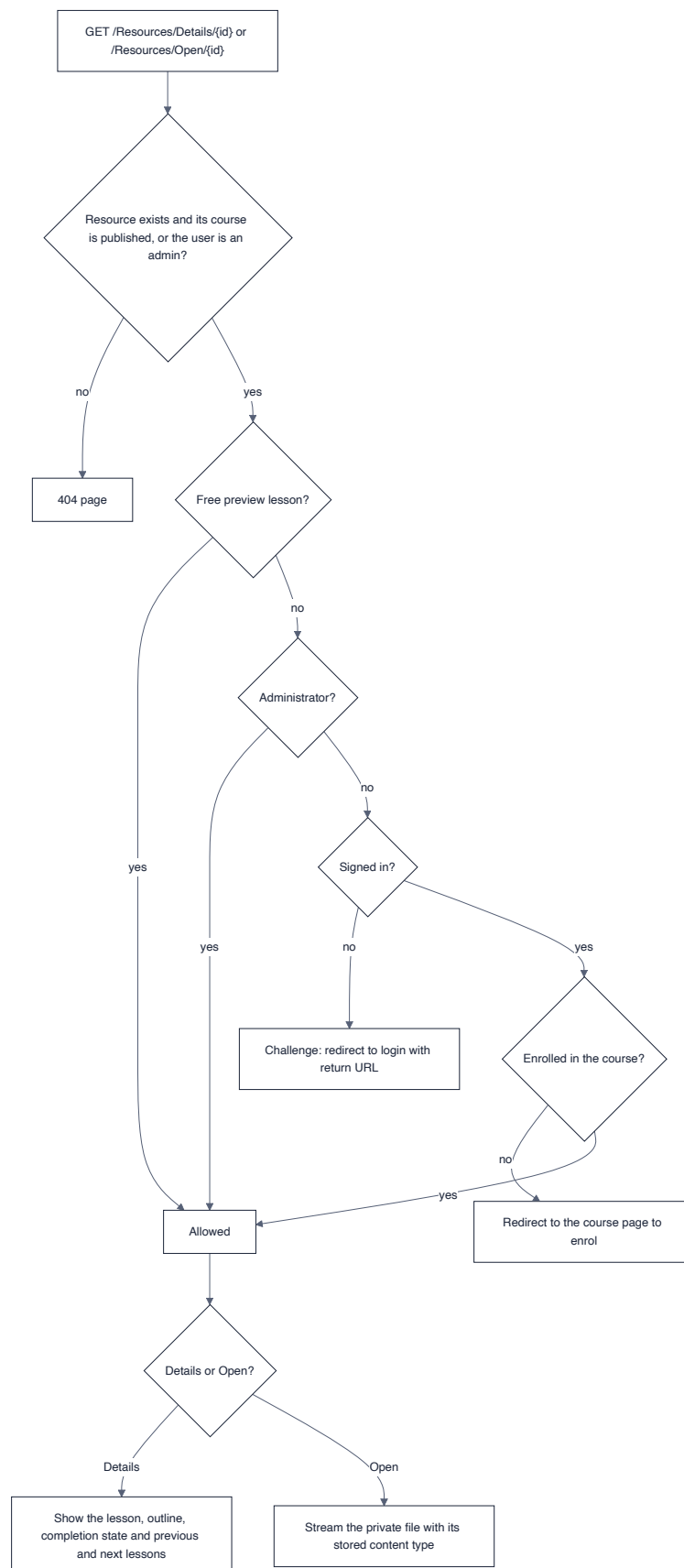
5. Enrolling in a course

`Controllers/CoursesController.cs` (Enroll) and `Services/EnrollmentService.cs`.



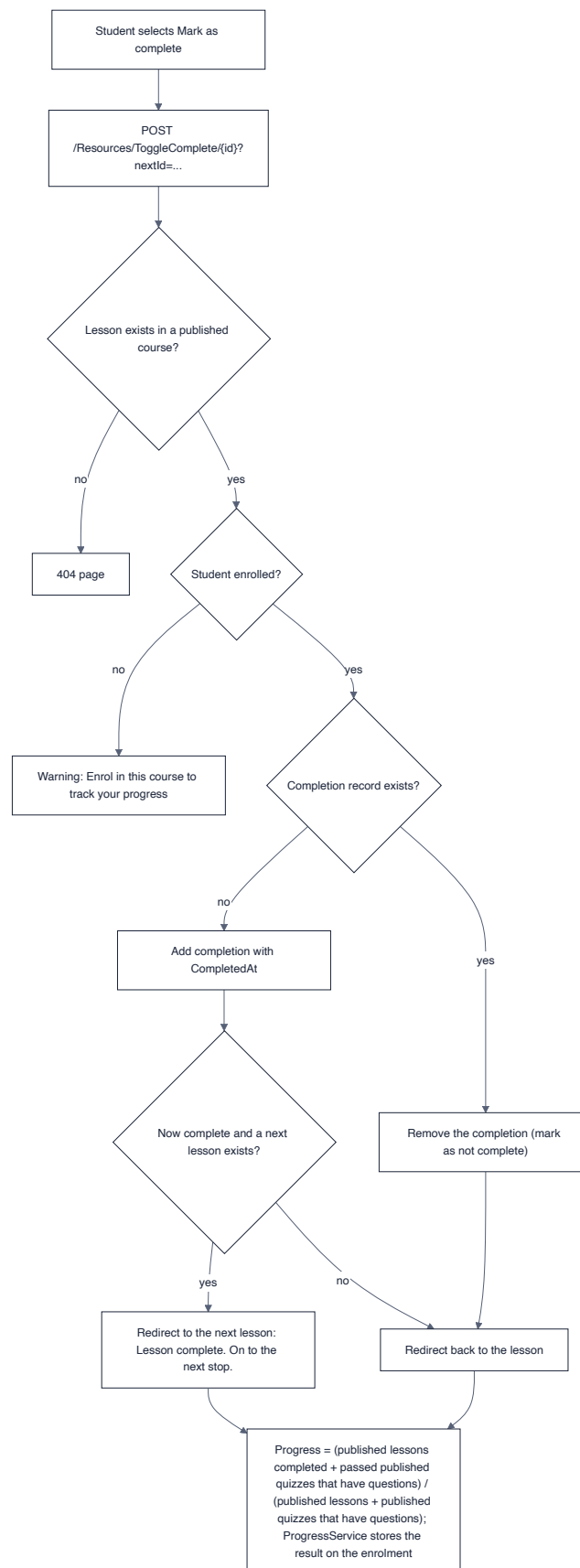
6. Opening a lesson or a lesson file

`Controllers/ResourcesController.cs` (Details, Open) and `Services/LearningResourceService.cs`.



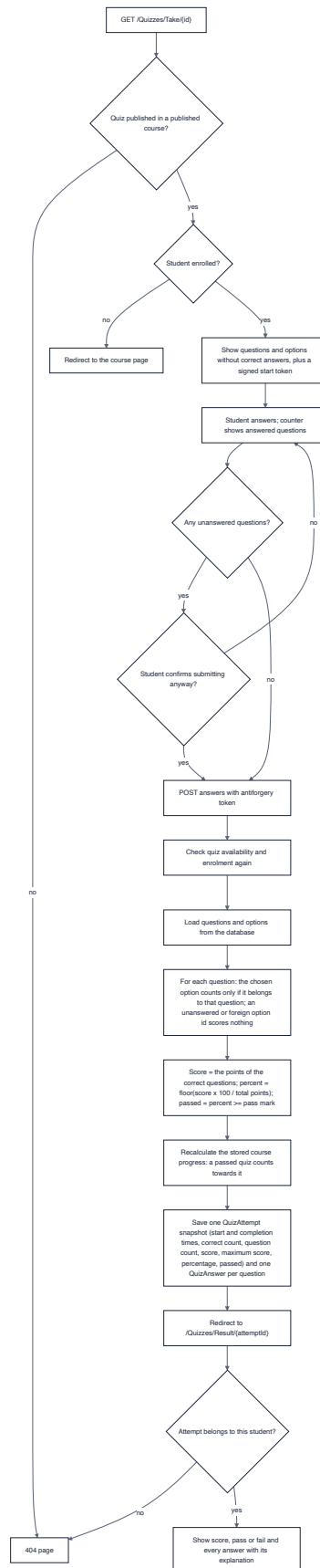
7. Marking a lesson complete and calculating progress

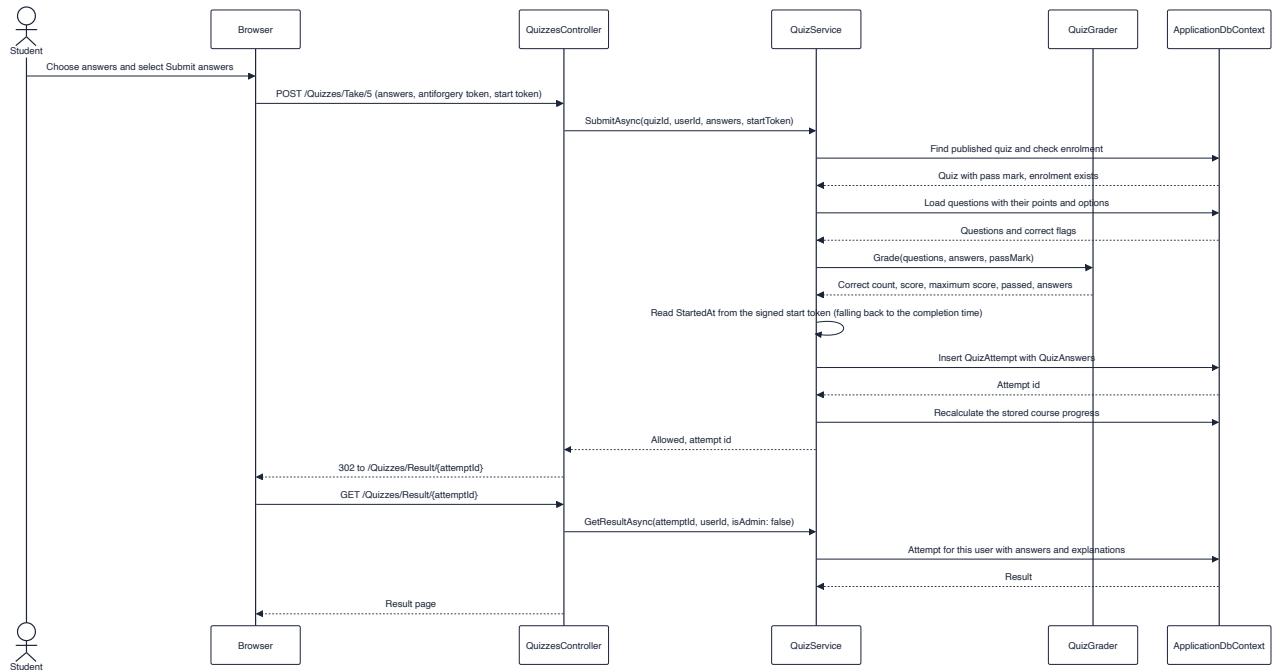
`Controllers/ResourcesController.cs` (ToggleComplete), `Services/LearningResourceService.cs` and `Services/ProgressService.cs`.



8. Taking and grading a quiz

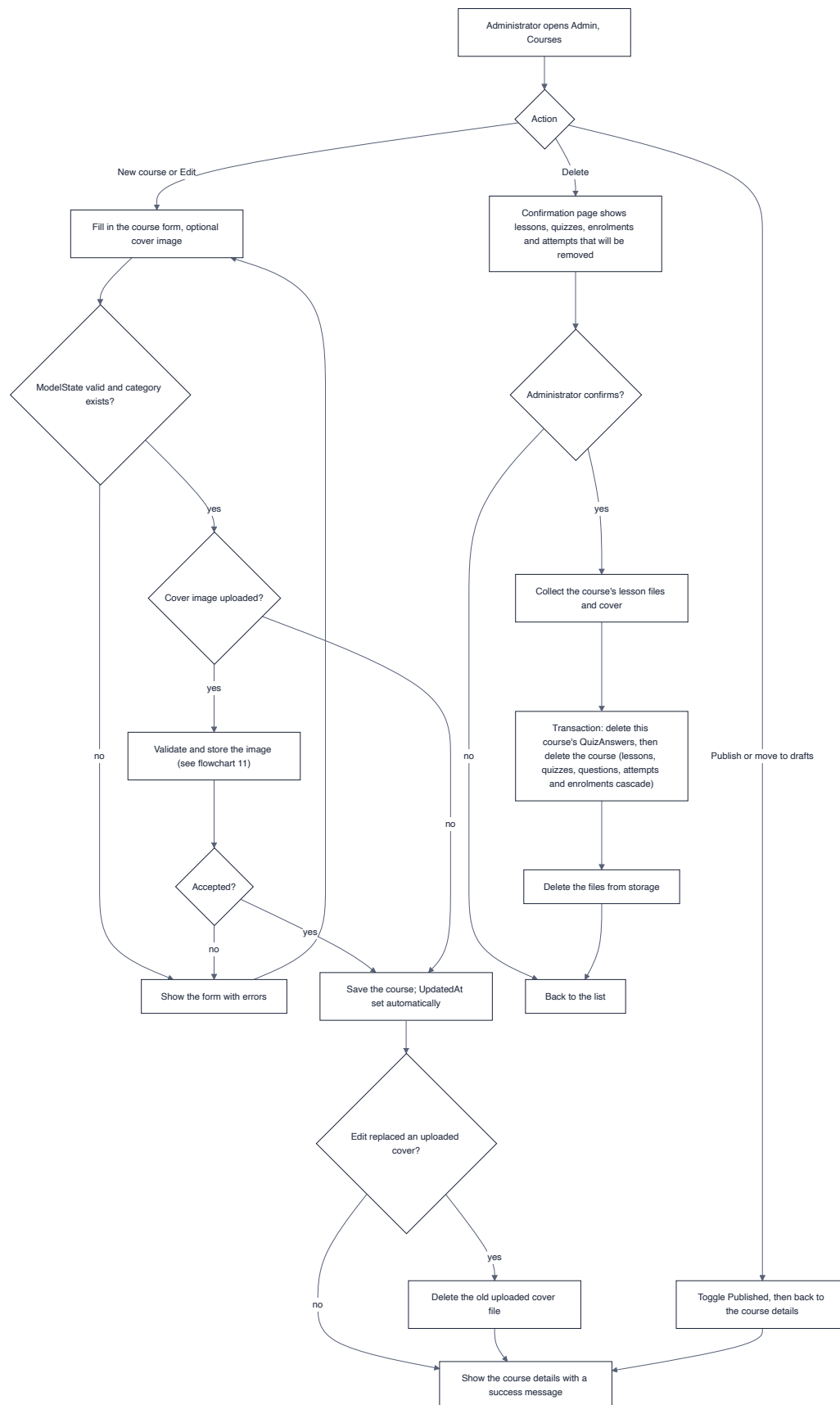
`Controllers/QuizzesController.cs` (Take, Result), `Services/QuizService.cs` and `Services/QuizGrader.cs`.





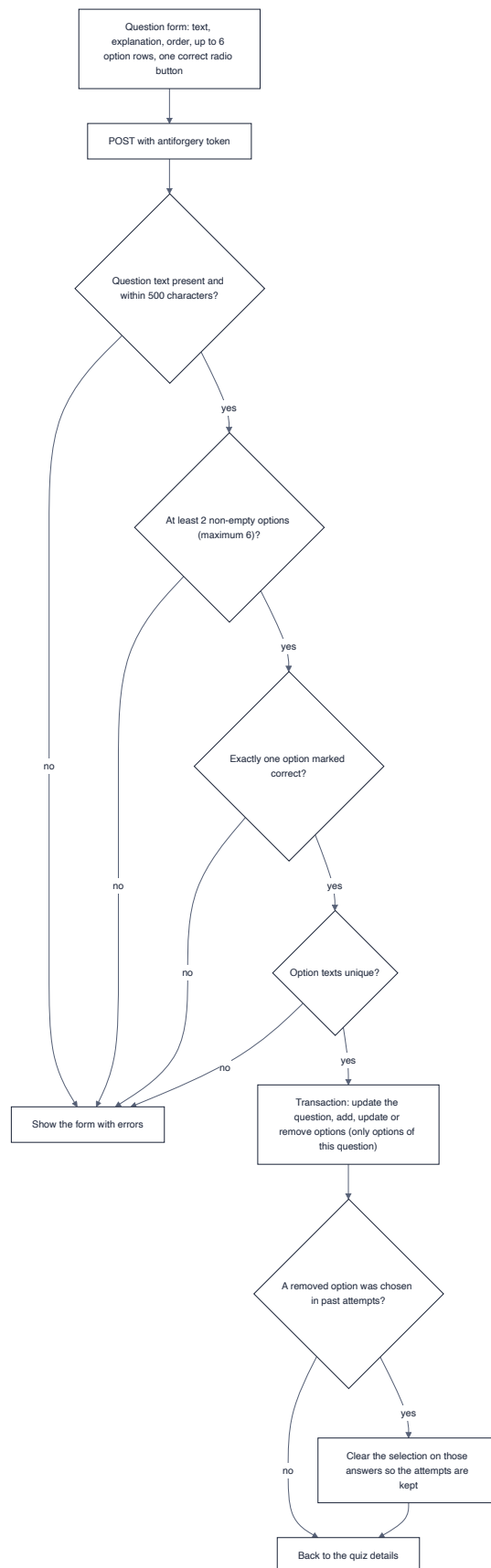
9. Admin: creating, editing and deleting a course

`Areas/Admin/Controllers/CoursesController.cs` and `Services/CourseManagementService.cs`.



10. Admin: saving a question

Areas/Admin/Controllers/QuestionsController.cs, ViewModels/Admin/AdminLearningViewModels.cs and Services/QuestionManagementService.cs.



11. File upload validation

`Services/Storage/FileStorageService.cs` and `Services/Storage/UploadRules.cs`. For learning resources, `Services/ResourceManagementService.cs` first checks that the file suits the resource type (a

PDF for a PDF resource, an image for an image resource).

